IOP Simulation Study − Population 1 (High IOP, omega_r = 0.357)

Figure 1 (Objective iii) − Normal QQ Plot of Standardised IOP Estimates

$z = \frac{\hat{\omega}_r - \omega_r}{SE_{JK}}$ | $\omega_r = 0.3571$ | $n_s = 475$ clusters/strat | k = 4 HH/cluster | B = 5000 reps

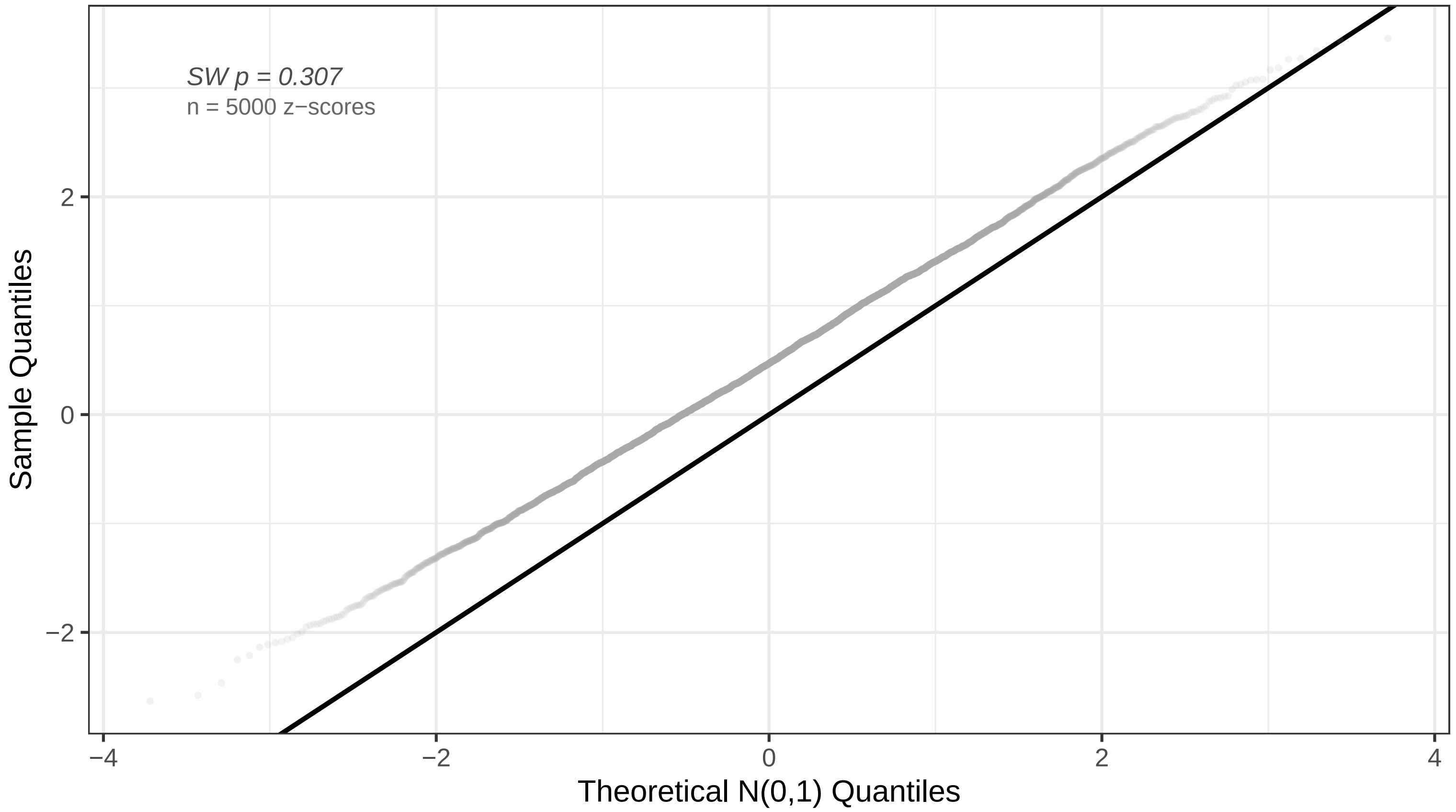

Figure 2 (Objective i) − Sampling Distribution of the Relative IOP Estimator

Bias = 0.00739 | RMSE = 0.01604 | MC SD = 0.01424 | B = 5000 reps

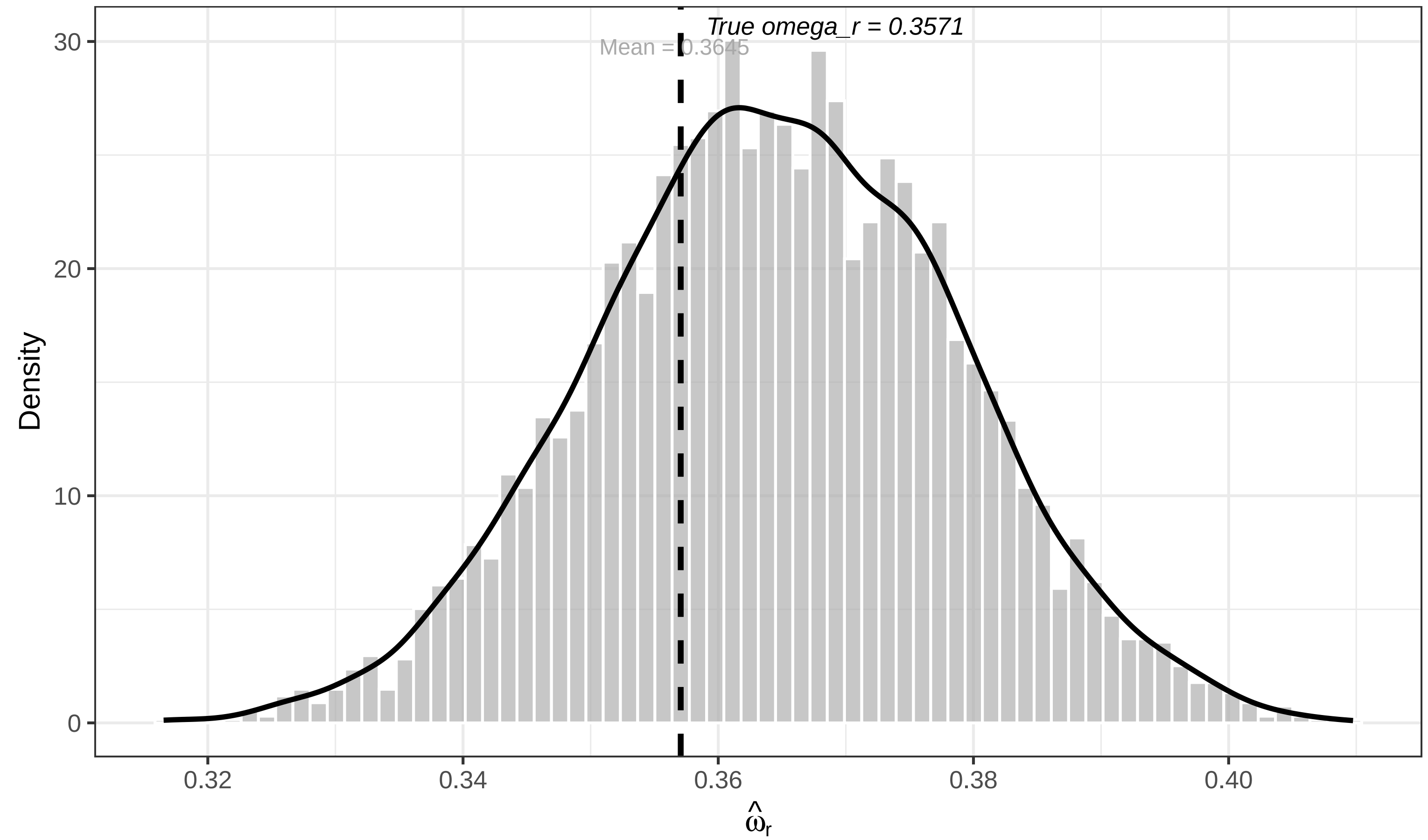

Figure 3 (Objective ii) − Empirical Coverage of 95% Wald Confidence Intervals

B = 5000 reps | $n_s = 475$ clusters/strat | k = 4 HH/cluster

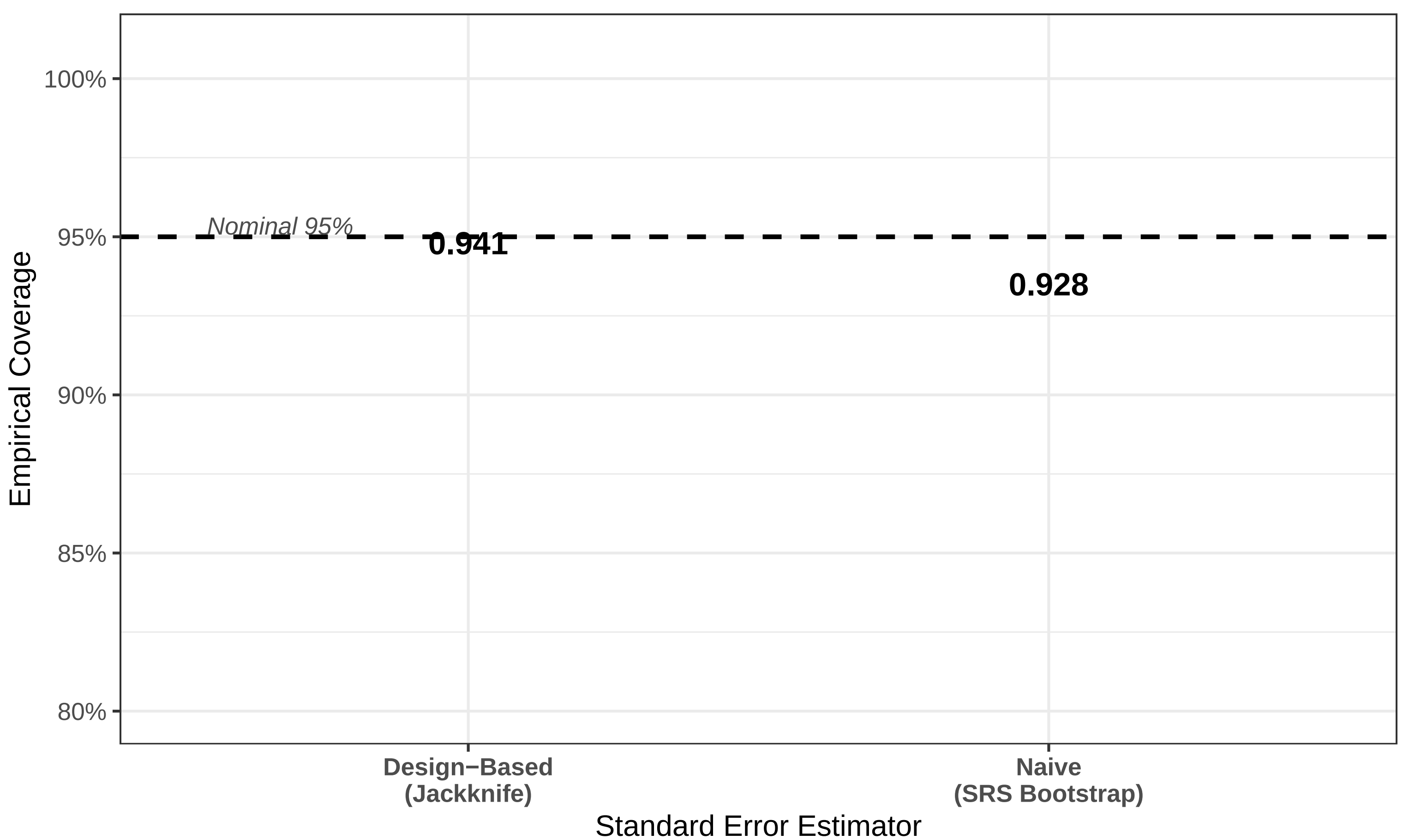

Figure 4 (Objective iv) − Type I Error and Power of the Two−Sample Wald Test

$H_0 : \omega_r^{(1)} = \omega_r^{(2)}$ vs $H_1 : \omega_r^{(1)} \neq \omega_r^{(2)}$ | $\alpha = 0.05$ | $n_s = 475$ clusters/strat per sample | B = 2000 reps